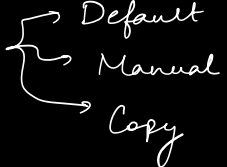


Today's Agenda :-

1) Access Modifiers

2) Constructors 

3) Destructor

4) Deep copy vs Shallow copy

5) Pass by val vs Pass by reference.

→ class

- 1) Store attributes & behaviour together of an idea ✓
- 2) Protect the attributes and behaviour
from illegitimate access from outside.

↳ Next class.

Types of Access Modifiers :-

1) public : Anyone can access

2) private : Nobody can access
Not even a child class
can accessible from the same class.

3) protected : Accessed only from the classes
in same package/folder.

4) default

	same class	Package	child class same pack	child diff pac	Anyone
Private	✓	✗	✗	✗	✗
Default	✓	✓	✓	✗	✗
Protected	✓	✓	✓	✓	✗
Public	✓	✓	✓	✓	✓

Class :- Blueprint of an idea / entity

Object :- Instance of the class

Student {

_____ }
}

int a = 12;

Student st = new Student();

create a
new object

Constructor

Student {

String name
int age = 20 ✓
double psp
String uniName
_____ }
}

void doSomething () {

_____ }

→ Automatically
generated

Student () {

name = NULL
age = 0
psp = 0.0
uniName = NULL
_____ }
}

Default Constructor :- If we don't create our

own constructor of class, a default constructor will be created.

The Default constructor creates a new object that sets the values of each attribute as the default value of that type.

1) Takes no params

2) No Return type

3) same name as class name

4) Sets every attribute to its default value

unless we have given some values.

5) Created only if we don't have our own constructor.

Custom/ Manual Constructor:-

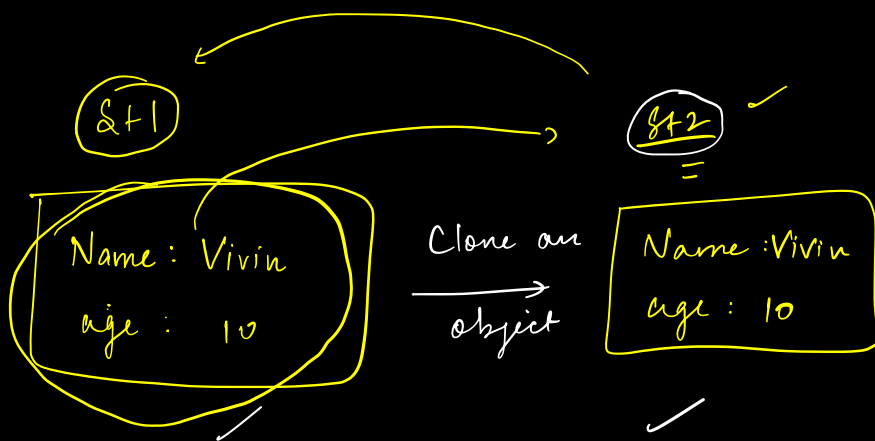
How a custom constructor works?

- 1) It initializes the object with the default values of every attribute
- 2) starts executing the lines inside the constructor.

Copy Constructor ::

We already have an object of Student

We want to create a new object of Student with exact same values as older object.



new Student (↓);

Student {

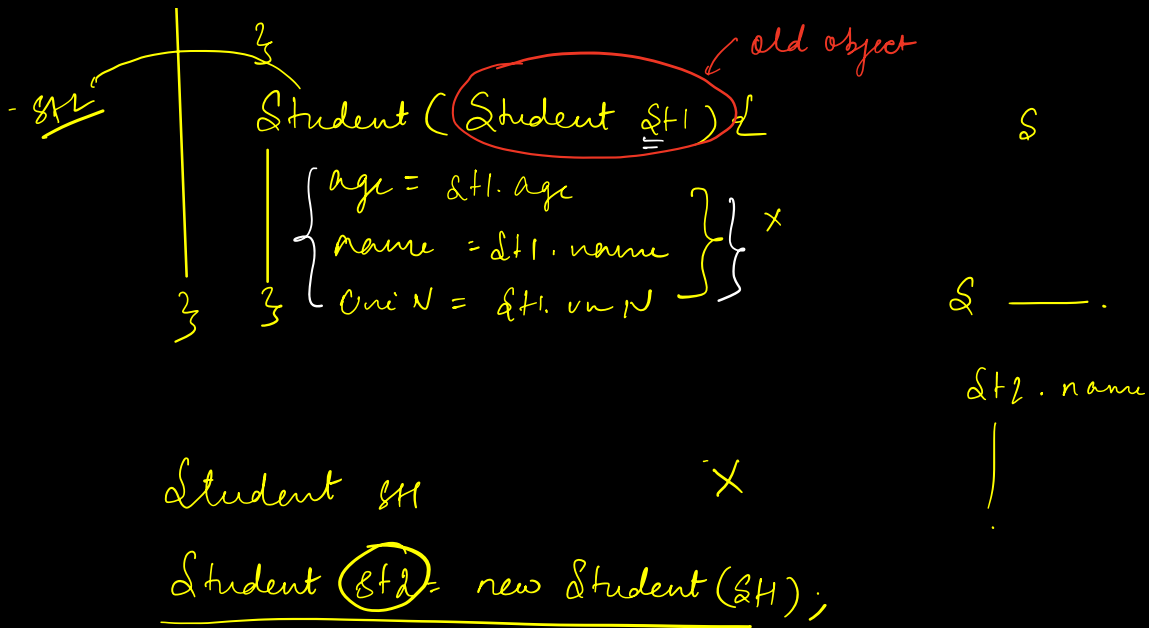
age ✓

name ✓

Vin Name ✓

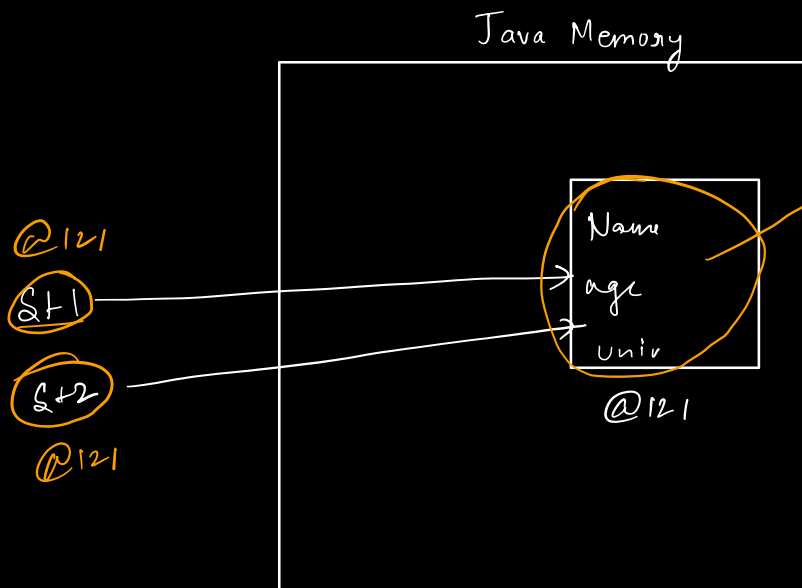
Student () {

|



`Student s2 = s1;`

You don't have a new object.



Types of data & where are they stored in memory?

1) Primitive Types | int | double | float

↳ as variables & their values directly

2) Objects

↳ Objects are stored in memory

Variables store the address of the

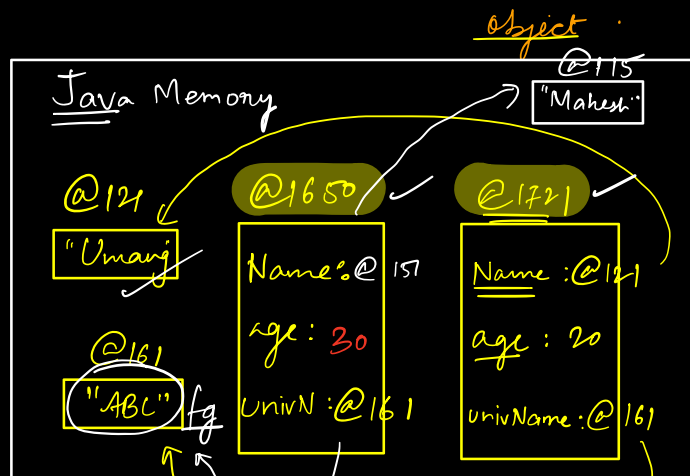
Variables

int a = 10

double p = 98.

Student s1 @1721

Student s2 = @1650



s2.age = 25

s2.name = "Mahesh"

abcfg
s2.UnivName add("fg")

s2.name = +

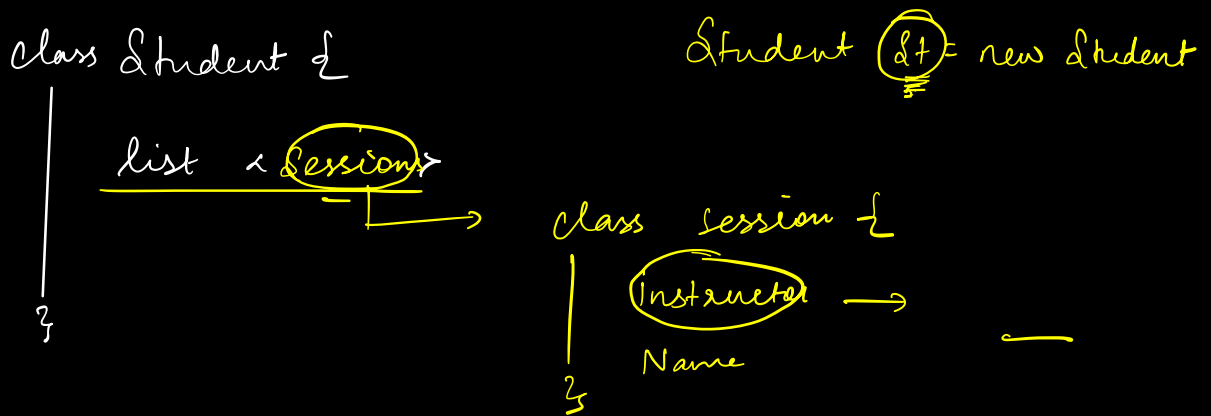
Deep copy vs Shallow Copy :-

Shallow Copy :- where I have created a new object but behind the scenes the new object still refers to a few attributes of older object.

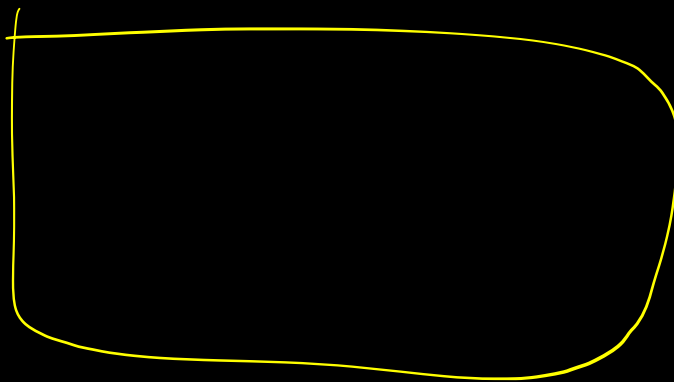
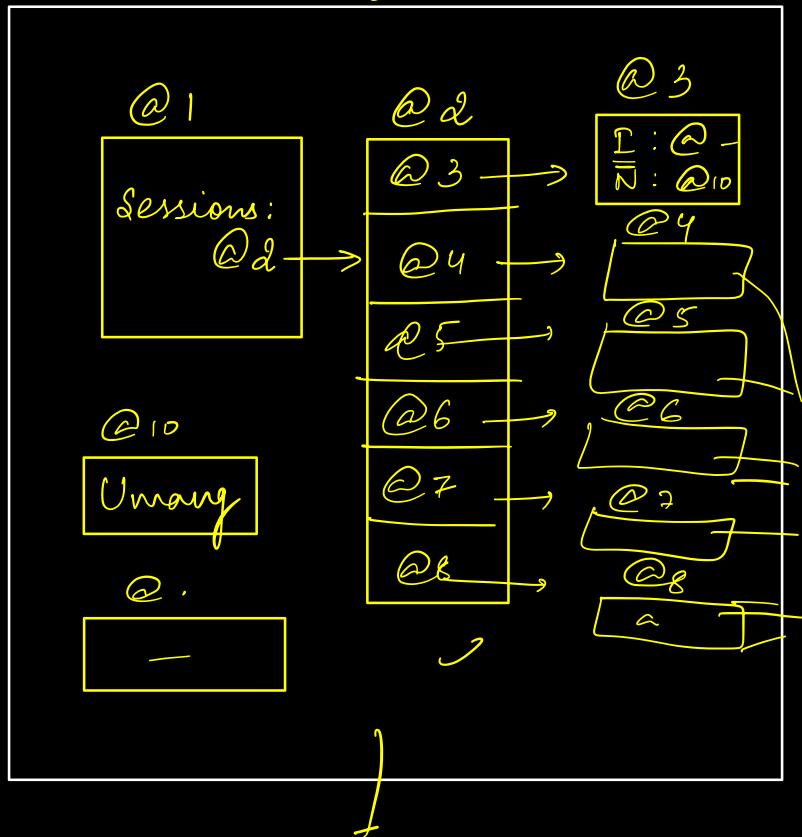
New & old object still have same data.

Deep Copy :- NO SHARED DATA

In Java, it is almost practically impossible to create a deep copy.



st @1



st+1

age	-
psp	-
b-id	-

↓

st+2

--

class st

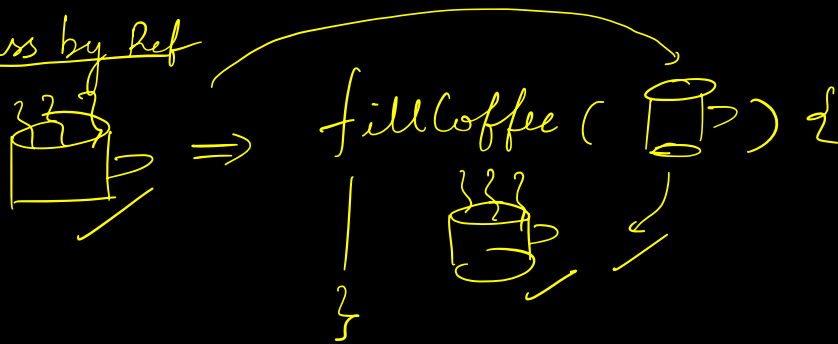
```

{
  age
  psp
  b-id
}

```

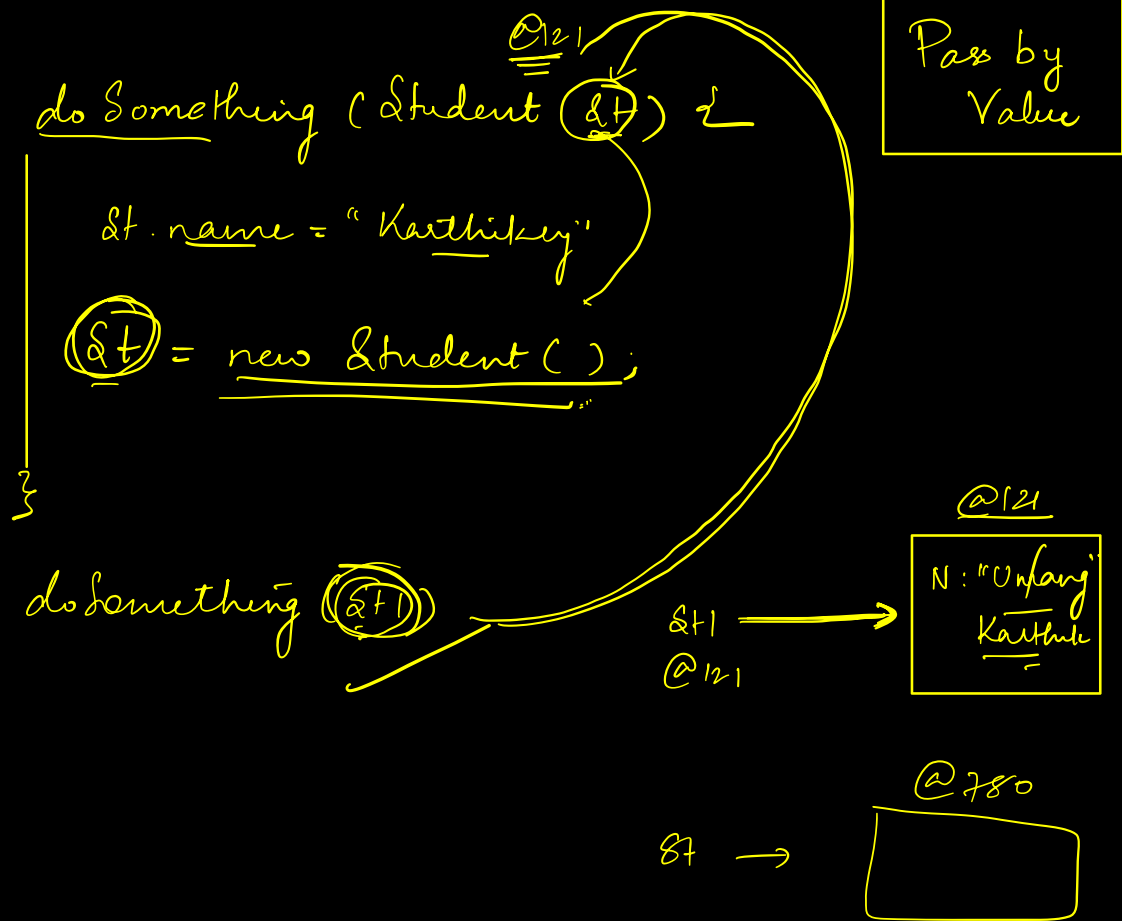
Is Java Pass by Value / Pass by Ref ?

Pass by Ref



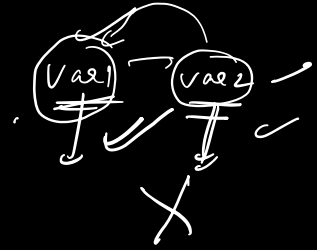
Pass by val





Summary

How object gets created !



⇒ Constructor.

↳ initialize all attributes with default value

→ run the code inside the constructor

→ work & use the object.

Once the obj is no longer needed,

(in)

Java by default gets them garbage collected.

Destructor

→ opposite of constructor

→ will be automatically called if object is
no longer needed to free
up the space used by
the object.

Next class ⇒ Inheritance }
Polymorphism }